Joshua Su

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Summary

Motivated Electrical Engineering student passionate about embedded systems and signal processing, particularly in autonomous systems and health monitoring.

Education

Georgia Institute of Technology — August 2024 – Expected May 2027
Bachelors of Science in Electrical Engineering, GPA: 3.88 — *Atlanta, GA*

Skills

Engineering | Embedded Systems (STM32, ESP32, & Arduino MCU's), Signal Processing, PCB Design (KiCAD) and Production, Circuit Design, Biosignals, Health Monitoring, Medical Image Processing

Programming & Software | Python, C/C++, MATLAB, , STM32CubeMX & IDE, OpenCV, PyTorch, Tensorflow, Android Studio, Machine Learning, Stream Processing, Data Visualization

Experience

Software Developer, Electrical Engineer — GT Medical Robotics Club — August 2025 – Present
Signal Processing, PCB Design | Python, KiCAD

- Designing glasses with an onboard ESP32 and Electrooculography (EOG) electrodes to act as a human-computer interface for individuals with locked-in syndrome
- Developed software (Python) to control cursor movement and clicks via eye tracking and blink detection, using ADCs, op-amps, low-pass filters, and signal differencing to acquire clean, usable signals
- Designing a custom microcontroller PCB with integrated filtering, analog conditioning, and signal processing for EOG glasses

Analog Mixed-Signal Design Engineer — SiliconJackets — March 2026 – Present
Transistor-level ASIC Design | Cadence Virtuoso

- Designed and layed out a custom ring oscillator given specific capacitor load, frequency range, power draw, and temperature requirements, and passed all DRC/LVS checks for onboarding project
- Designed a frequency divider module and phase error detector module for a phase-locked-loop (PLL)
- PLL will be integrated with other analog modules and be used by the digital design team with the overall goal of taping out a functioning RISC-V processor

Research Intern — Emory University (The Kamaleswaran Lab) — June – December 2023
Signal Processing, Computer Vision | Python — *Atlanta, GA*

- Analyzed 400+ lung CT-scans and successfully employed 2 computer-vision models for early Interstitial Lung Disease classification, achieving 96% accuracy and saving hours of manual labeling time
- Gained experience with medical image processing in 2D and 3D (windowing, spatial filtering, segmentation, and image reconstruction), and machine learning
- Collaborated with multidisciplinary teams and presented research findings

Projects

Embedded Crash Detection Device — December 2025
Embedded Systems | ESP32, C/C++

- Engineered a wearable/mountable crash and fall detection device using ESP32, MPU6050 accelerometer, and NEO-6M GPS module, capable of autonomously identifying high-impact events and immediately transmitting the user's real-time location to emergency contacts.
- Designed to reduce emergency response times by automating crash detection, eliminating delays caused by manual distress calls.
- Gained hands-on experience programming microcontrollers and implementing real-time sensor algorithms to distinguish high-impact events from normal motion using accelerometer data thresholds.

Relevant Coursework

Intro to Signal Processing, Digital System Design, Circuit Analysis, Probability and Statistics with Applications, Differential Equations